```
import pandas as pd
```

```
#read csv
df = pd.read_csv(r"/content/House-Prices.csv")
```

```
print(df.columns)
```

```
Index(['ID', 'Date', 'Price', 'Bedrooms', 'Bathrooms', 'Sqft_living',
       'Sqft_lot', 'Floors', 'Waterfront', 'View', 'Condition', 'Grade',
       'Sqft_above', 'Sqft_basement', 'Yr_built', 'Yr_renovated', 'zipcode',
       'Lat', 'Long', 'Sqft_living15', 'Sqft_lot15'],
      dtype='object')
```

```
df.tail()
```

	ID	Date	Price	Bedrooms	Bathrooms	Sqft_living	Sqft_
21608	21609	20140725T000000	365000.0	5	2.00	1600	4
21609	21610	20150311T000000	380000.0	2	1.00	1040	7
21610	21611	20140624T000000	339000.0	3	1.00	1100	4
21611	21612	20140703T000000	399900.0	2	1.75	1410	1
21612	21613	20141030T000000	268950.0	3	1.00	1320	8

5 rows × 21 columns

```
df.shape
```

```
(21613, 21)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21613 entries, 0 to 21612
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   ID             21613 non-null  int64
 1   Date           21613 non-null  object
 2   Price          21613 non-null  float64
 3   Bedrooms       21613 non-null  int64
 4   Bathrooms      21613 non-null  float64
 5   Sqft_living    21613 non-null  int64
 6   Sqft_lot       21613 non-null  int64
 7   Floors         21613 non-null  float64
```

```
 8   Waterfront     21613 non-null  int64
 9   View           21613 non-null  int64
 10  Condition      21613 non-null  int64
 11  Grade          21613 non-null  int64
 12  Sqft_above     21613 non-null  int64
 13  Sqft_basement  21613 non-null  int64
 14  Yr_built       21613 non-null  int64
 15  Yr_renovated   21613 non-null  int64
 16  zipcode        21613 non-null  int64
 17  Lat            21613 non-null  float64
 18  Long           21613 non-null  float64
 19  Sqft_living15  21613 non-null  int64
 20  Sqft_lot15     21613 non-null  int64
dtypes: float64(5), int64(15), object(1)
memory usage: 3.5+ MB
```

```
df.describe()
```

	ID	Price	Bedrooms	Bathrooms	Sqft_living	Sq
count	21613.00000	2.161300e+04	21613.000000	21613.000000	21613.000000	2.1613
mean	10807.00000	5.401822e+05	3.370842	2.114757	2079.899736	1.5106
std	6239.28002	3.673622e+05	0.930062	0.770163	918.440897	4.1420
min	1.00000	7.500000e+04	0.000000	0.000000	290.000000	5.2000
25%	5404.00000	3.219500e+05	3.000000	1.750000	1427.000000	5.0400
50%	10807.00000	4.500000e+05	3.000000	2.250000	1910.000000	7.6180
75%	16210.00000	6.450000e+05	4.000000	2.500000	2550.000000	1.0688
max	21613.00000	7.700000e+06	33.000000	8.000000	13540.000000	1.6513

```
df.duplicated()
```

	0
0	False
1	False
2	False
3	False
4	False
...	...
21608	False
21609	False
21610	False
21611	False
21612	False

21613 rows × 1 columns

dtype: bool

```
df.isnull().sum()
```

	0
ID	0
Date	0
Price	0
Bedrooms	0
Bathrooms	0
Sqft_living	0
Sqft_lot	0
Floors	0
Waterfront	0
View	0
Condition	0
Grade	0
Sqft_above	0
Sqft_basement	0
Yr_built	0
Yr_renovated	0
zipcode	0
Lat	0
Long	0
Sqft_living15	0
Sqft_lot15	0

dtype: int64

```
#count missing values in price column
print(df["Price"].isna().sum())
```

```
0
```

```
median_value = df["Price"].median()
print(median_value)
```

```
450000.0
```

```
df["Price"] = df["Price"].fillna(median_value)
```

```
print(df["Price"].isna().sum())
```

```
0
```

```
df.isnull().sum()
```

	0
ID	0
Date	0
Price	0
Bedrooms	0
Bathrooms	0
Sqft_living	0
Sqft_lot	0
Floors	0
Waterfront	0
View	0
Condition	0
Grade	0
Sqft_above	0
Sqft_basement	0
Yr_built	0
Yr_renovated	0
zipcode	0
Lat	0
Long	0
Sqft_living15	0
Sqft_lot15	0

dtype: int64

```
import pandas as pd
# The 'Description' column does not exist in this dataset.
```

```
# Removing the failing dropna line and checking current null counts.
print(df.isnull().sum())
```

```
ID               0
Date             0
Price            0
Bedrooms         0
Bathrooms        0
Sqft_living      0
Sqft_lot         0
Floors           0
Waterfront       0
View             0
Condition        0
Grade            0
Sqft_above       0
Sqft_basement    0
Yr_built         0
Yr_renovated     0
zipcode          0
Lat              0
Long             0
Sqft_living15    0
Sqft_lot15       0
dtype: int64
```

```
print('Checking for missing values in existing columns:')
print(df.isnull().sum())
```

```
Checking for missing values in existing columns:
ID               0
Date             0
Price            0
Bedrooms         0
Bathrooms        0
Sqft_living      0
Sqft_lot         0
Floors           0
Waterfront       0
View             0
Condition        0
Grade            0
Sqft_above       0
Sqft_basement    0
Yr_built         0
Yr_renovated     0
zipcode          0
Lat              0
Long             0
Sqft_living15    0
Sqft_lot15       0
dtype: int64
```

```
print('Final confirmation of no missing values:')
print(df.isnull().sum())
```

```
Final confirmation of no missing values:
ID               0
Date             0
Price            0
Bedrooms         0
Bathrooms        0
Sqft_living      0
Sqft_lot         0
Floors           0
Waterfront       0
View             0
Condition        0
Grade            0
Sqft_above       0
Sqft_basement    0
Yr_built         0
Yr_renovated     0
zipcode          0
Lat              0
Long             0
Sqft_living15    0
Sqft_lot15       0
dtype: int64
```

```
df.isnull().sum()
```

	0
ID	0
Date	0
Price	0
Bedrooms	0
Bathrooms	0
Sqft_living	0
Sqft_lot	0
Floors	0
Waterfront	0
View	0
Condition	0
Grade	0
Sqft_above	0
Sqft_basement	0
Yr_built	0
Yr_renovated	0
zipcode	0
Lat	0
Long	0
Sqft_living15	0
Sqft_lot15	0

dtype: int64

```
q1= df["Price"].quantile(0.25)
q3= df["Price"].quantile(0.75)
iqr = q3-q1
lower_bound = q1 - 1.5*iqr
upper_bound = q3 + 1.5*iqr
outliers = df[(df["Price"] < lower_bound) | (df["Price"] > upper_bound)]
print(outliers)
```

```
          ID             Date      Price  Bedrooms  Bathrooms  Sqft_living  \
50        51  20140911T000000  1490000.0         5       3.50         3620
72        73  20150221T000000  1510000.0         5       3.00         3300
```

```
73        74  20140630T000000  1730000.0         5       3.50         5000
74        75  20140617T000000  2030000.0         4       3.75         4100
109      110  20150324T000000  1430000.0         4       5.00         4630
...      ...              ...        ...       ...        ...          ...
21561  21562  20141118T000000  1390000.0         3       1.25         2400
21566  21567  20140806T000000  2250000.0         3       3.25         3890
21567  21568  20150227T000000  1450000.0         4       3.50         2860
21570  21571  20150304T000000  2480000.0         5       3.75         6810
21571  21572  20141006T000000  1440000.0         4       3.00         3720

       Sqft_lot  Floors  Waterfront  View  ...  Grade  Sqft_above  \
50         7821     2.0           0     2  ...     10        2790
72        33474     1.0           0     3  ...      9        1870
73        26540     2.0           0     3  ...     10        3410
74        22798     1.5           0     3  ...     11        2540
109       24054     2.0           0     3  ...     11        4630
...         ...     ...         ...   ...  ...    ...         ...
21561      6653     3.0           0     2  ...     11        2400
21566      3452     2.0           0     0  ...     12        2890
21567      2199     3.0           0     0  ...     10        2860
21570      7500     2.5           0     0  ...     13        6110
21571      5000     2.5           0     0  ...      9        2720

       Sqft_basement  Yr_built  Yr_renovated  zipcode      Lat     Long  \
50               830      1958          2010    98040  47.5738 -122.215
72              1430      1957          1991    98040  47.5673 -122.210
73              1590      2008             0    98040  47.5665 -122.210
74              1560      1934          1979    98040  47.5648 -122.210
109                0      2005             0    98075  47.5870 -121.966
...              ...       ...           ...      ...      ...      ...
21561              0      1992             0    98122  47.6019 -122.290
21566           1000      2006             0    98102  47.6260 -122.323
21567              0      2013             0    98102  47.6262 -122.323
21570            700      1922             0    98102  47.6285 -122.322
21571           1000      1905             0    98102  47.6282 -122.322

       Sqft_living15  Sqft_lot15
50              2690        9757
72              3836       20953
73              3360       17398
74              3880       18730
109             4630       17584
...              ...         ...
21561           1910        6653
21566           2860        2199
21567           1990        1378
21570           2660        7500
21571           2610        5000

[1159 rows x 21 columns]
```

```
import matplotlib.pyplot as plt
```